

Parent Guide — primary / module-09-subtraction-nikhilam

IKS Curriculum

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Parent Guide — Module 9: Friends to Ten! (Towards *Nikhilam*)

Your child's age: Primary (Grades 3–5) · **Module length:** 10 sessions × 35 min

What your child is learning

A foundational mental-math skill called “**friends to ten**” — knowing instantly which digit completes another to make 10:

- $1 + 9 = 10$
- $2 + 8 = 10$
- $3 + 7 = 10$
- $4 + 6 = 10$
- $5 + 5 = 10$

And then a step up: using “friends to ten” to instantly subtract from 100. ($100 - 7 = 93$, because 7's friend is 3, and we shift one over.)

At the end of the module they'll meet the Sanskrit name *Nikhilam* — pronounced “ni-khi-lam” — and a small magic phrase: “all from nine and the last from ten.” They won't learn the full method until Grade 6, but they'll see it once and be impressed.

What to expect this fortnight

A daily chant:

“1 and 9 are friends! 2 and 8 are friends! 3 and 7 are friends! 4 and 6 are friends! 5 and 5 are friends! All make TEN!”

Games at home: - A card-based “Friends to Ten” game (Day 1). - Quick-fire friend-naming drills with a parent or sibling. - A “coin game” (Day 5): your child plays shopkeeper; you hand them 100-rupee paper money for a small item; they give change.

A creative project (Days 7–10): Your child will design their own “Friends to Ten” game — cards, board, dice, jumping — and play classmates' games on Day 10.

How to help — and how NOT to help

Helpful: - 5-minute friend-drill before bed: call out digits; your child says the friend. Try to beat their time. - Be the customer in the coin game: you hand a 100; for a 67-rupee item, your child gives change of 33. They say “67’s friend is 33!” - For their project: provide materials (cards, paper, dice) and let *them* design the rules.

Not helpful: - Insisting that “the way I was taught” is the only right way. The “complete to 10” trick gives the same answer as standard borrowing. - Designing their game for them. - Anxiety about speed. Speed comes from practice, not pressure.

A note on “Vedic mathematics”

You may have heard the term “Vedic mathematics.” Here is the honest framing we use:

- The “friends to ten” rule with the Sanskrit name *Nikhilam* is a real, useful Indian mental-arithmetic technique.
- The full rule comes from a book by Bharati Krishna Tirthaji called *Vedic Mathematics* (1965).
- Tirthaji claimed the rule comes from a Vedic appendix-text. **No other Sanskritist has confirmed this.** The method is real; the Vedic attribution is unverified.
- We teach the method enthusiastically AND are honest about its history. By Senior band (Grade 9–12) your child will read the scholarly debate.

What success looks like

By the end of the module, your child can: - Recall all 5 friends-to-ten instantly. - Compute $100 - N$ mentally for small N (under 20). - Pronounce and recognise the word *Nikhilam*. - Show classmates a game they designed.

Want to go deeper?

The Middle-band version (Grade 6–8) teaches the full Nikhilam method, including subtractions from 1000 and 10 000, and connects it to how computers subtract in binary. Preview: curriculum/middle/module-09-subtraction-nikhilam/README.md.

10-Day Lesson Plan

Module 9 (Primary) — 10-Day Lesson Plan

At a glance (35-min sessions)

Day	Title	Activity	Output
1	Friends to Ten	Story + chant + Make-Ten card game	First-day fun
2	Hugs to Ten	Pair drill	Each child names all 5

Day	Title	Activity	Output
			friends
3	The Magic Word	Introduce <i>Nikhilam</i> via story	Chant the word 5×
4	100 minus Friends	$100 - 7 = 93$, bar-model	Drawing
5	The Coin Game	Shopkeeper play	Game card
6	All from 9, Last from 10	Gentle rule, with hand actions	Mini-poster
7	Big-Friends Day	Friends-to-100 of 2-digit numbers	Worksheet
8	Project Day 1	Game design	Sketch + first prototype
9	Project Day 2	Build + test	Game ready to play
10	Game Showcase	Play each other's games	Stickers

Daily structure (35 min)

Time	What
5 min	Settle + the “Friends to Ten” chant
8 min	Story / new game introduction
17 min	Game / activity
5 min	Sharing / reset

The daily chant

“1 and 9 are friends! 2 and 8 are friends! 3 and 7 are friends! 4 and 6 are friends! 5 and 5 are friends! All make TEN!”

Sing it / chorus it with claps. By Day 3 it should be effortless. Same chant as Module 5; children come in with it half-learned.

Pacing notes

- Games > worksheets at this band. If a worksheet feels slow, drop it and add another game round.
- Day 3 (the *Nikhilam* magic word) is a “wow” moment — keep it short and mysterious.
- Day 6 introduces “all from 9, last from 10” — but only verbally, with hand actions. Don’t push for full mastery; that’s a Middle-band goal.
- Days 8–10 build to the game showcase. Confirm groups by Day 6.

What children leave with

- Instant recall of all 5 friends-to-ten.
- A mental habit of “completing” when subtracting.
- The Sanskrit word *Nikhilam* — pronounced and recognised.
- One game they made and one they played.

Homework

Homework — Subtraction — Nikhilam (All from 9) (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Sūtra — All from 9, Last from 10** · Draw about today's concept. (~5 min)
2. **Day 2 — Complement to 1000 and Beyond** · Draw about today's concept. (~5 min)
3. **Day 3 — Subtract from a Power of 10** · Draw about today's concept. (~5 min)
4. **Day 4 — General Subtraction via Complements** · Draw about today's concept. (~5 min)
5. **Day 5 — Why It Works — The Algebra** · Draw about today's concept. (~5 min)
6. **Day 6 — Bar Model Proof** · Draw about today's concept. (~5 min)
7. **Day 7 — Two's Complement in Computers** · Draw about today's concept. (~5 min)
8. **Day 8 — Speed Day** · Draw about today's concept. (~5 min)
9. **Day 9 — Mixed Practice** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.